

Homework Exercise Sheet

Ex. 6 Solution

Label (S) : only one correct answer; (M) : more than one correct answer

1. (S) Compute DFT of $[2, 2 + i, 1 - i, -3 + 2i]$.

- a) $[2 + 2i, -4i, 4 - 4i, 2 + 6i]$.
- b) $[1 + 4i, 5 - 4i, 3 - 6i, 3 + 6i]$.
- c) $[5 + 2i, -1 + i, -5, 5 - 2i]$.
- d) $[6 + 2i, 0, -4, 6 - 2i]$.

Solution:

a). Use formula 4.2.1.19 from the lecture document.

2. (S) Compute inverse DFT of $[1, 2, 3, 4]$.

- a) $\frac{1}{2} [5, -1 + i, -1, -1 + i]$.
- b) $\frac{1}{2} [5, -1 + i, -1, -1 - i]$.
- c) $\frac{1}{2} [5, -1 - i, -1, -1 + i]$.
- d) $\frac{1}{2} [5, -1 - i, -1, -1 - i]$.

Solution:

c). Use formula 4.2.1.20 from the lecture document.

3. (S) Find the result of applying the LT-FIR filter with the impulse response $h = [1, 2, 3, 4]$ to the sequence $x = [5, 6, 7, 8]$.

- a) $[5, 16, 34, 60]$.
- b) $[60, 61, 52, 32]$.
- c) $[32, 52, 61, 60, 34, 16, 5]$.
- d) $[5, 16, 34, 60, 61, 52, 32]$.

Solution:

d). Compute the convolution, $y_k = \sum_j h_k x_{k-j}$, for all integer values of j and k . For out-of-range indices of h and x , set their values to zero. Provide only nonzero values of y as the result.

4. (S) What is the output of an LT-FIR filter with impulse response $(\dots, 0, 1, 2, \dots, 100, 0, \dots)$, if the input is a constant signal $(x_j)_{j \in \mathbb{Z}}, x_j = 1$?

- a) $(\dots, 0, 1, 2, \dots, 100, 0, \dots)$
- b) $(\dots, 0, 1, 1, \dots, 1, 0, \dots)$
- c) $(\dots, 1, 1, 1, \dots, 1, 1, \dots)$
- d) $(\dots, 5050, 5050, 5050, \dots, 5050, 5050, \dots)$

Solution:

d). Due to the properties of LT-FIR, the response at position k can be decomposed into combinations of impulse response and signal (Lecture Note 4.1.2.4): $F((x_j)_{j \in \mathbb{Z}})_k = \sum_{j=0}^{100} h_j x_{k-j} = \sum_{j=0}^{100} h_j = 5050$.

5. **(S)** Given a LT-FIR filter with impulse response $\mathbf{h} = (\dots, 0, 4, 2, 1, 0, \dots)$ and an output signal of $\mathbf{y} = (\dots, 0, 16, 32, 16, 22, 8, 4, 0, \dots)$, what is the input signal $\mathbf{x}$ (truncated to integer)?
- a) $(\dots, 0, 4, 6, 0 \dots)$
 - b) $(\dots, 0, 4, 6, 0, 4, 0 \dots)$
 - c) $(\dots, 0, 4, -5, 0, 0, 20, -40, 20, 0 \dots)$
 - d) $(\dots, 0, 4, -5, 0, 0, 20, -40, 20, -10, 0 \dots)$

Solution:

b). Applying each answer to the filter, you can find that b) gives the right output signal. To solve it, the filtering can be seen as a convolution $\mathbf{h} * \mathbf{x} = \mathbf{y}$ which in turn can be seen as doing a matrix multiplication $\mathbf{A}_\mathbf{h} \mathbf{x} = \mathbf{y}$ where $\mathbf{A}_\mathbf{h} \in \mathbb{R}^{6 \times 6}$ is a Toeplitz matrix defined by the filter $\mathbf{h}$ (See Lecture Note 4.1.3.2). The filter matrix has to be extended to size 6×6 to match the size of the output. Then solving the linear equation $\mathbf{A}_\mathbf{h} \mathbf{x} = \mathbf{y}$ gives the result. Note that an output signal of finite length from LT-FIR does not necessarily come from an input signal of finite length as the inverse of a banded lower triangular Toeplitz normally results is not necessarily a banded lower triangular matrix (it is still lower triangular though).

6. **(M)** Denote by $(\mathbf{x}_k) \in \mathbb{K}^k$ a sequence $[x_0, \dots, x_{k-1}]$. Which of the following statements regarding discrete convolution (denoted by $*$) and discrete periodic convolution (denoted by $*_n$ with period n) is **wrong**?
- a) $(\mathbf{x}_m) * (\mathbf{y}_n) \in \mathbb{K}^{m+n}$.
 - b) $(\mathbf{x}_n) *_n (\mathbf{y}_n) \in \mathbb{K}^{2n}$.
 - c) $(\mathbf{x}_m) * (\mathbf{y}_n) = (\mathbf{y}_n) * (\mathbf{x}_m)$
 - d) $(\mathbf{x}_n) *_n (\mathbf{y}_n) = (\mathbf{y}_n) *_n (\mathbf{x}_n)$
 - e) $(\mathbf{x}_n) * (\mathbf{x}_n) = \mathbf{0}$ implies that $(\mathbf{x}_n) = \mathbf{0}$.
 - f) $(\mathbf{x}_n) *_n (\mathbf{x}_n) = \mathbf{0}$ implies that $(\mathbf{x}_n) = \mathbf{0}$.
 - g) $(\mathbf{x}_m) * (\mathbf{y}_n) = \mathbf{0}$ implies that either $(\mathbf{x}_m) = \mathbf{0}$ or $(\mathbf{y}_n) = \mathbf{0}$.
 - h) $(\mathbf{x}_n) *_n (\mathbf{y}_n) = \mathbf{0}$ implies that either $(\mathbf{x}_n) = \mathbf{0}$ or $(\mathbf{y}_n) = \mathbf{0}$.

Solution:

a), b), g), h). The corrected statement of a) is $(\mathbf{x}_m) * (\mathbf{y}_n) \in \mathbb{K}^{m+n-1}$, see Definition 4.1.3.4. For discrete periodic convolution, the resulting vector has the same size as the convoluted vectors, which gives the corrected statement of b). The commutativity of discrete convolution is proved in Remark 4.1.3.5, while the one for periodic convolution follows the same proof. To prove Statements e) and f), one can resort to the convolution theorem (Theorem. 4.2.2.2) $(\mathbf{u}) * _n (\mathbf{x}) = \mathbf{F}_n^{-1}[(\mathbf{F}_n \mathbf{u})_j (\mathbf{F}_n \mathbf{x})_j]_{j=1}^n$. Since Fourier matrices are regular and thus $\mathcal{N}(\mathbf{F}_n) = \emptyset$, we can conclude that $(\mathbf{x}_n)$ must be zero. The same argument can be applied to (non-periodic) discrete convolution by conversion to periodic convolution. Statement h) can also be disproved via the convolution theorem. As long as vectors at frequency domain $(\mathbf{F}_n \mathbf{x})$ and $(\mathbf{F}_n \mathbf{y})$ are orthogonal, two non-zero signals, whose convolution is vanishing, can be constructed. Statement g) follows by conversion to periodic convolution.

7. **(S)** Given two vectors $\mathbf{h} \in \mathbb{K}^n, \mathbf{x} \in \mathbb{K}^m$, their convolution can be computed via periodic convolution as follows

$$\mathbf{h} * \mathbf{x} = (\tilde{\mathbf{h}} *_N \tilde{\mathbf{x}})_{0:p},$$

where $\tilde{\mathbf{h}} := [\mathbf{h}, \mathbf{0}]^\top \in \mathbb{K}^N$ and $\tilde{\mathbf{x}} := [\mathbf{x}, \mathbf{0}]^\top \in \mathbb{K}^N$ are zero-padded vectors. Please specify the minimal vector length N that makes the computation valid and the corresponding index range $0:p$ (C++ indexing, e.g., start from index 0).

- a) $N = n + m, \quad p = n + m - 1$
- b) $N = n + m - 1, \quad p = n + m - 2$
- c) $N = n + m - 2, \quad p = n + m - 3$
- d) $N = \max\{n, m\}, \quad p = \max\{n, m\} - 1$
- e) $N = \max\{n, m\}, \quad p = n + m - 2$
- f) $N = 2\max\{n, m\}, \quad p = 2\max\{n, m\} - 1$
- g) $N = 2\max\{n, m\}, \quad p = n + m - 2$

Solution:

b). Here we provide a rigorous solution but some intuition of how convolution works may be very straightforward to yield the answer. We can rewrite the discrete convolution of two vectors (C++ indexing used)

$$(\mathbf{h} * \mathbf{x})_k = \sum_{j=\max\{0, k-n+1\}}^{\min\{m-1, k\}} (\mathbf{x})_j (\mathbf{h})_{k-j}, \quad k = 0, \dots, m+n-2,$$

Next, we closely examine the periodic convolution of these two vectors, exploiting our knowledge about zero components of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{h}}$:

$$\begin{aligned} (\tilde{\mathbf{h}} *_N \tilde{\mathbf{x}})_k &= \sum_{j=0}^{N-1} (\tilde{\mathbf{x}})_j (\tilde{\mathbf{h}})_{(k-j) \bmod N} = \sum_{j=0}^{m-1} (\mathbf{x})_j (\tilde{\mathbf{h}})_{(k-j) \bmod N} \\ &= \sum_{j=0}^{\min\{m-1, k\}} (\mathbf{x})_j (\tilde{\mathbf{h}})_{k-j} + \sum_{j=\min\{m, k+1\}}^{m-1} (\mathbf{x})_j (\tilde{\mathbf{h}})_{k+N-j}. \end{aligned}$$

- Case $k \in \{m-1, \dots, N-1\}$: In this case the second sum becomes void and we find

$$(\tilde{\mathbf{h}} *_N \tilde{\mathbf{x}})_k = \sum_{j=0}^{m-1} (\mathbf{x})_j (\tilde{\mathbf{h}})_{k-j} = \sum_{j=\max\{0, k-n+1\}}^{\min\{m-1, k\}} (\mathbf{x})_j (\mathbf{h})_{k-j} = (\mathbf{h} * \mathbf{x})_k,$$

that is, for these vector components both discrete convolutions produce the same result for any $N \geq \max\{m, n\}$.

- Case $k \in \{0, \dots, m-2\}$: The corresponding components of the discrete periodic convolution are

$$\begin{aligned} (\tilde{\mathbf{h}} *_N \tilde{\mathbf{x}})_k &= \sum_{j=0}^k (\mathbf{x})_j (\tilde{\mathbf{h}})_{k-j} + \sum_{j=k+1}^{m-1} (\mathbf{x})_j (\tilde{\mathbf{h}})_{k+N-j} \\ &= \sum_{j=\max\{0, k-n+1\}}^k (\mathbf{x})_j (\mathbf{h})_{k-j} + \sum_{j=k+1}^{m-1} (\mathbf{x})_j (\tilde{\mathbf{h}})_{k+N-j} \end{aligned}$$

Clearly, the second sum should be evaluated to zero if both discrete convolutions are to yield the same vector components. Thus, we have to demand

$$\begin{aligned} (\tilde{\mathbf{h}})_{k+N-j} &= 0 & \forall j \in \{k+1, \dots, m-1\}, \quad \forall k \in \{0, \dots, m-2\} \\ \Leftrightarrow k+N-j &\geq n & \forall j \in \{k+1, \dots, m-1\}, \quad \forall k \in \{0, \dots, m-2\} \\ \Leftrightarrow N-(m-1) &\geq n \\ \Leftrightarrow N &\geq n+m-1, \end{aligned}$$

Hence, $N \geq n+m-1$ is a necessary and sufficient condition for the agreement of the vector components with indices $k \in \{0, \dots, m-2\}$.

Clearly, the range of the resulting sequence has to be $0 : n+m-2$ since $\mathbf{h} * \mathbf{x} \in \mathbb{K}^{n+m-1}$.

8. (S) Discrete periodic convolution of two vectors can be computed via discrete Fourier transform. The following shows two functions that aim to compute discrete periodic convolution. Please select the correct statement.

```
1 Eigen::VectorXcd pconvfft1(const Eigen::VectorXcd &u,
2                             const Eigen::VectorXcd &x) {
3     Eigen::FFT<double> fft;
4     return fft.inv(((fft.fwd(u)).cwiseProduct(fft.fwd(x))).eval());
5 }

1 Eigen::VectorXcd pconvfft2(const Eigen::VectorXcd &u,
2                             const Eigen::VectorXcd &x) {
3     Eigen::FFT<double> fft;
4     const int n = u.size();
5     return n * fft.fwd(((fft.inv(u)).cwiseProduct(fft.inv(x))).eval());
6 }
```

- a) pconvfft1 is correct while pconvfft2 is wrong.
- b) pconvfft1 is wrong while pconvfft2 is correct.
- c) Both functions are wrong.

d) Both functions are correct.

Solution:

a). `pconvfft1` is the standard implementation provided in Code 4.2.2.4. `pconvfft2` needs a bit more scrutiny. Provided the fact that $\mathbf{F}_n = \frac{1}{n}\overline{\mathbf{F}}$, we know that `pconvfft2` is actually calculating $\frac{1}{n}\mathbf{F}(\overline{\mathbf{F}}\mathbf{u} \cdot \overline{\mathbf{F}}\mathbf{x})$ which is in general different from the correct output $\mathbf{u} * \mathbf{x} = \mathbf{F}^{-1}(\mathbf{F}\mathbf{u} \cdot \mathbf{F}\mathbf{x}) = \frac{1}{n}\overline{\mathbf{F}}(\mathbf{F}\mathbf{u} \cdot \mathbf{F}\mathbf{x})$.